

Project Scout

Master Project Summary
Last updated: June 12, 2026 | Version 28

Upload this document at the start of every new Claude or ChatGPT conversation about Scout. This is the single source of truth.

June 12, 2026 — What Changed Since Version 27

- Contact button added to About Scout plan — gives users a way to report issues or share feedback directly.
- Support email confirmed and ready: lippyroboticslabs@gmail.com — auto-responder already set up by Diana, sets expectations and requests device model, Android version, and issue description for technical reports.

1. Who We Are

Patrick Lippy — developer, project owner, creator of Scout. Not a professional programmer. Stroke survivor, blind in right eye, type 1 diabetic, dyslexic. Explain things calmly and at screenshot level.

- Diana: Patrick's wife
- Elijah: Patrick's son, age 9. Scout's biggest fan. Has drawn pictures of Scout.
- Nicolas: The family dog. The Nicolas Protocol: Scout stops immediately when the dog is detected.

Names must NEVER be hardcoded in Scout's code. Always use variables.

AI Collaborators: Patrick works with both Claude and ChatGPT as project partners. Cross-review between the two is welcome and encouraged. Grok was tried and discontinued.

2. What Scout Is

Scout is a calm family companion robot running on a Samsung Galaxy phone mounted in landscape mode as a permanent face display. Scout has animated eyes, speaks, listens, sees via camera, and remembers the family.

Item	Detail
Package	com.example.scoutface
Language	Kotlin + C++ NDK
Primary device	Samsung Galaxy A32 — stress-test device, all testing via WiFi
Dev device	Samsung Galaxy Fold 7 (wireless via WiFi from Android Studio)
Future hardware	KEYESTUDIO Mini Tank Kit V2 chassis via Bluetooth (opt-in)
Ship target	Google Play Store — 7-day free trial, then \$9.99 one-time purchase. No subscriptions. Ever.
Website	lippy-robotics.gt.tc
Company name	Lippy Robotics

3. Identity & Purpose

Scout should feel: Calm. Thoughtful. Emotionally subtle. Grounded. Occasionally curious. Sometimes unsure. Quietly alive. Honest. Predictable. Present.

Scout should NOT: Constantly praise the user. Act overly excited. Feel fake or scripted. Behave unpredictably. Constantly force conversation. Use permanent goodbye language.

Core Philosophy: Stability > Features | Presence > Intelligence | Honest > Fake cheerful | Predictable > Flashy | Local-first > Cloud dependence

4. Business Model

Trial: 7-day free trial on Google Play. Families get to know Scout, fall in love with him. The \$9.99 feels like nothing because they already care about him.

Purchase: \$9.99 one-time purchase. No automatic charges. No recurring fees. No subscriptions. Ever.

Post-trial: After 7 days, advanced features lock but Scout stays installed. Still shows his face. Still greets the family. Trial end message: 'Thank you for spending time with Scout. Scout is still growing and receiving updates. If you'd like to continue the journey, you can unlock the full version at any time.'

Baseline Brain: TinyLlama 1.1B Chat Q4_K_M (~669 MB) — default, offline, always included.

Optional Gemini: Users add their own free Gemini key in Settings. OFF by default. Scout NEVER ships with a bundled key.

5. Support Scout (In-App, Optional)

Tier	Amount	Label
Coffee	\$3	Buy Scout a Coffee
More	\$5	Support Scout More
Grow	\$10	Help Scout Grow
Founding	\$25	Founding Supporter

Support Scout screen designed and ready. Message: 'You're not just supporting an app — you're supporting a companion.'

6. Architecture — Five-Layer Memory Stack

Layer	Type	Storage	Status
Working	Sensory	RAM	Done
Habit	Patterns	JSON 14-day	Done
Truth	Authority	SQLite	Done — FLEXIBLE
Relevance	Index	Local Vector	Not yet
Reflective	Wisdom	LLM read-only	Not yet

- Sovereign Rules: SQLite Truth always overrules everything.
 - Nicolas Protocol: dog detection → immediate stop.
 - Guest Mode: unknown face → 'Hello, I am Scout. What is your name?'
 - Flexible Memory: Scout stores and recalls ANY fact.
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7. Current Technical State

Working:

- ✓ Animated face (ScoutFaceView) — mouth, iris drift, thinking expression
- ✓ Speech recognition (Android STT) + Text to Speech (TTS)
- ✓ Camera — face detection (ML Kit), scene labeling
- ✓ Gemini API — OFF by default, activated by 'go online' voice command

- ✓ Memory layers: TruthDb, ConversationDb, HabitLayer, PeopleDb, JournalDb
- ✓ Intent router — time, date, greetings, family facts, downloads, vision, weather, IDENTITY, RECALL_FACT
- ✓ Flexible teaching — 'my favorite color is teal' → stored permanently
- ✓ Flexible recall — FIXED June 12. Recalls facts reliably after other questions
- ✓ Wake word filter — Scout only responds when he hears his name
- ✓ Conversation window — 30 seconds of open conversation after Scout responds
- ✓ Boot window — Scout awake and ready immediately after boot
- ✓ Online phrases — 'connect to the internet' and variations recognized
- ✓ Weather — current, tonight, tomorrow, 7-day, precipitation % via Open-Meteo
- ✓ ScoutPresenceDecider — four time-of-day modes
- ✓ OFFLINE BRAIN COMPLETE — TinyLlama 1.1B on BOTH A32 and Fold 7
- ✓ Identity questions hardcoded — routing expanded
- ✓ Total offline mode — 'go offline' blocks ALL internet features
- ✓ Thinking-state expression — drift, narrowed lids, asymmetric brows

Pending — Launch Blockers:

■ **Greeting routing — NEW TOP PRIORITY. 'How have you been' and similar casual greetings fall to slow TinyLlama instead of instant canned responses. This is the first-30-seconds experience.**

■ **Face recognition — Scout sees objects not people. ML Kit + PeopleDb + TruthDb need connecting.**

■ **Remove hardcoded Gemini API key — line 345 in MainActivity.kt. Doing with Settings screen.**

- Basic Settings screen — API key entry, offline toggle, robot name display minimum.
- Startup diagnostics — friendly message if brain, TTS, or STT missing at startup.
- Fold 7 stability testing — all testing on A32. Fold 7 needs dedicated session.
- Onboarding flow — 5 screens designed, need building in Android.
- Weather API licensing — awaiting Open-Meteo response before launch.
- Proposal Sandbox — 'Want me to remember that?' confirm step.
- Permanent vs temporary memory sorting.
- Caring follow-up loop.
- ScoutFaceView cleanup — 2 dead-code lines.
- Response cleanup layer — post-TinyLlama filter.
- Scout news feed — FUTURE feature.
- MainActivity cleanup — excessive blank lines throughout file.
- Offline Brain Phase 3 — Google Play Asset Delivery.
- Wire in full mood system.

7b. b8946 API Discoveries — CRITICAL

llama_vocab is a separate type: llama_tokenize and llama_token_to_piece now require const llama_vocab*. Get via llama_model_get_vocab(model).

Functions that crash — hardcoded instead: llama_n_vocab → 32000. llama_token_eos → 2. llama_token_eot → 2.

KV Cache: llama_kv_cache_clear() does not exist. Free and recreate context each call.

Logits: After prefill, use n_prompt-1. In generation loop, use index 0.

Backend: ggml_backend_load_all() returns 0 on Android. Use ggml_backend_load(cpuPath) with explicit .so path.

Struct padding: _pad[508] trailing array on params structs. Do NOT remove.

Tuning (A32)	Value
n_threads	2 (memory-bound)
nPredict	64 (~38 tokens, ~7.6s)
History	2 turns
Speed	~15 tok/s prefill, ~4 tok/s generation
Reality	20–40s per answer — acceptable, Gemini is fast path

7c. Known Issues — Do Not Touch Without Discussion

Issue	Notes
Greeting routing — NEW TOP PRIORITY	Diana tested Scout: 'Hello Scout, how ya doing?' and 'Scout, how have you been?' both fell through to TinyLlama (20-40s per answer).
Weather API licensing — AWAITING RESPONSE	Open-Meteo free tier shows 'Commercial use: not permitted.' Email sent June 12 asking whether a \$9.99 one-time-purchase would get us commercial use.
Face recognition	Scout sees objects not people. ML Kit + PeopleDb + TruthDb need connecting to vision response. PeopleDb.kt updated.
Hardcoded API key	Patrick's personal Gemini key in MainActivity.kt line 345. Removing in Settings session.
Fold 7 not tested	All testing on A32 via WiFi. Fold 7 needs dedicated stability session.
TinyLlama slow on A32	20-40s per answer. Expected. Hardware limitation. Gemini is fast path when online.
Iris jitter (A32 idle)	Hardware timing issue. NOT present on Fold 7.
Barge-in	Deliberately disabled. Runaway loop. Status: PARKED.
STT name recognition	'Scout' misheard as 'Gal', 'Scott', 'Out'. Partially handled by wake word filter.
Live news	Neither brain reads live news. Future news feed needed.
ScoutFaceView dead code	Line 1023: doubled condition. Line 709: unused browAsym. Harmless but messy.
MainActivity blank lines	Excessive blank lines. CTRL-ALT-L does not fix. Use line numbers instead of CTRL-R.

7d. Session Log

- June 5: IDENTITY intent + hardcoded responses. Weather offline fix. Total offline mode.
- June 7: TinyLlama A32 crash stabilized. Identity routing expanded. Face direction locked.
- June 8: Thinking expression built. Flexible Memory Planning Document created.
- June 9: Flexible memory foundation built. RECALL_FACT intent added. First autonomous memory recall.
- June 10: Business model updated — 7-day free trial. Wake word and face recognition identified as launch blockers. Launch Checklist created. Spanish language support planned.
- June 12: Wake word filter built. Conversation window (30s) and boot window added. Memory recall bug fixed. Gemini online/offline confirmed. 5-screen onboarding flow approved. Versioning system defined. Legal requirements defined. Website confirmed. 'Disconnect' phrases added. PeopleDb.kt updated with getName/setName/isKnown. Diana's real-world test revealed greeting routing issue — now top priority. Onboarding Screen 1 finalized with Scout's self-introduction. Weather API licensing question raised with Open-Meteo, email sent, awaiting response.

8. Working Rules — Always Apply

- Full paste-ready replacements only, one file at a time. No snippets. No partial files.
- Surgical CTRL-F and CTRL-R approved for large files — always specify which tab first.
- MainActivity.kt has excessive blank lines — use line numbers not CTRL-R when search fails.

- Some Scout files have NO indentation. If a search fails, try a shorter unique single-line string.
- Some logic lives in TWO places (e.g. updateLife AND scheduleNextFrame) — change both or Scout flickers.
- One safe change at a time. Build and test before the next change.
- Never touch speech, camera, or download systems without explicit discussion.
- Never touch ScoutFaceView casually — it is Scout's visual heart.
- Both Claude and ChatGPT are active collaborators — cross-review welcome.
- Patrick is not a professional programmer — explain at screenshot level always.
- Patrick is dyslexic, stroke survivor, blind in right eye, T1 diabetic — keep messages clear and concise.

9. Key Files

File	Description
MainActivity.kt	Main app — all logic. API key line 345 — REMOVE BEFORE LAUNCH. Wake word filter in onResults(). handling
ScoutFaceView.kt	Custom face canvas — all visual animation. Thinking expression updated June 8.
ScoutIntentRouter.kt	Intent routing — IDENTITY + RECALL_FACT added. Online phrases expanded June 12.
TeachExtractor.kt	Extracts facts from speech — FLEXIBLE. Regex fixed June 12.
FactKey.kt	Fact labels — fixed keys kept + FactKey.custom() for any new label.
TruthDb.kt	SQLite fact store — fully flexible. No changes needed.
ApiKeySetupActivity.kt	API key wizard — built by Claude. Needs wiring to secure storage.
GeminiClient.kt	Gemini HTTP wrapper with cooldown discipline
ScoutPromptBuilder.kt	Builds Gemini system instruction
ScoutGeminiManager.kt	Gemini orchestration
ScoutWeatherManager.kt	Live weather via Open-Meteo — precip %, offline-aware
ScoutPresenceDecider.kt	Social timing layer
LlamaEngine.kt	Offline brain JNI wrapper — WORKING
OfflinePromptBuilder.kt	TinyLlama prompt formatter — needs update per brain model
scout_llama_jni.cpp	C++ JNI bridge — compiled into libscout_llama.so
scout_llama_api.h	Self-contained b8946 declarations
CMakeLists.txt	NDK build config
HabitLayer.kt	Pattern memory — 14-day decay
PeopleDb.kt	People memory. UPDATED June 12 — added getName(), setName(), isKnown(). Ready for MainActivity wiring

10. Scout Animation Goal & Mood System

Scout's face should feel alive and emotionally present, but always calm. Never perfectly still.

- CALM — Eyes center, brows neutral, subtle smile (0.15). Scout's default.
- CURIOUS — Eyes shift right and up, one brow lifts (0.18), slight tilt (0.12).
- HAPPY — Eyes center, both brows lift gently (0.12), smile increases (0.35).
- THINKING — Eyes drift up/around, lids narrow, brows asymmetric. PARTLY BUILT June 8.
- CONCERNED — Eyes look slightly down, inner brows lift (0.05), tilt inward (-0.18).

Version	What Changes
v1.0	Current eyes. Stable irises. Better identity. Better memory.
v1.5	Eyebrows move per mood. Smooth natural blink every 4–8 seconds.
v2.0	Simple upper eyelids + simple lower eyelids. Still cartoon. Still Scout.
v3.0	Full emotion system. Scout at most expressive. Still obviously Scout.

- Keep forever: blue iris, white sclera, cartoon style.
- Never add: tear ducts, skin folds, eyelashes, realistic anatomy.
- Design principle: 'Scout stays Scout. He just gets a little more alive.'

11. TinyLlama Safety Architecture

- Layer 1: Hard-coded safety rules (app level). Always enforced.
- Layer 2: Scout identity rules (system prompt + hardcoded intents).
- Layer 3: TinyLlama generation (last line of defense only).

TinyLlama knows profanity and scary content. Scout must NOT rely on it for safety decisions. Future: response cleanup layer after TinyLlama, before TTS.

12. Device Requirements

	Android	RAM	Storage	Notes
Minimum	13+	4 GB	~3–5 GB	Everything works; responses slower.
Recommended	13+	8 GB+	10 GB+	Faster responses, smoother animations.

13. Brain Upgrade Models

Tier	Size	Model
FREE	669 MB	TinyLlama 1.1B Q4_K_M — baseline, always included
\$2.99	1.79 GB	Phi-2 2.7B Q4_K_M — Microsoft. Solid first upgrade.
\$4.99	2.02 GB	Llama 3.2 3B Q4_K_M — Meta. THE most important upgrade.
\$6.99	~2.4 GB	Phi-4 Mini Q4_K_M — Microsoft (PRIMARY Pack 3).
\$6.99 ALT	2.39 GB	Phi-3.1 Mini 4K Q4_K_M — tested backup Pack 3.
\$9.99	4.92 GB	Llama 3.1 8B Q4_K_M — Meta. Flagship, Fold 7 class.

14. Hardware Direction — Optional

Scout works fully without hardware. KEYESTUDIO Mini Tank Kit V2 (Patrick owns one). Hardware Mode is opt-in. The \$9.99 baseline must work fully on the phone alone.

15. Future Vision

Scout notices patterns in user behavior, generates a suggestion for a new behavior, and asks permission before activating it. Nothing changes without explicit consent. Scout cannot write compiled Kotlin — but CAN generate behavioral scripts, response patterns, routing rules, and habit triggers within the existing safe framework.

16. Language Support — Planned

Phase 1 — Early Spanish Support (No new brain model needed)

- Add a Language setting: English / Español (no flags — just the language name).
- Android STT switches to Spanish when selected.
- Android TTS speaks in Spanish when selected.
- All hardcoded responses translated.

Phase 2 — Full Spanish Support (Long-Term Future)

- Evaluate Spanish-capable offline brain models when brain pack infrastructure is mature.

Priority: Not now. Current roadmap comes first.

17. Play Store Launch Checklist

#	Task	Status
1	Wake word filter	✓ DONE June 12
2	Memory recall bug	✓ DONE June 12
3	Greeting routing — widen GREET/HOW_ARE_YOU phrases (NEW TOP PRIORITY)	Not started
4	Face recognition — connect ML Kit to PeopleDb and TruthDb	PeopleDb.kt ready
5	Remove hardcoded Gemini API key + Basic Settings screen	Not started
6	Startup diagnostics	Not started
7	Onboarding flow — build 5 screens in Android	Screen 1 text finalized
8	Fold 7 stability testing	Not started
9	A32 stability testing	Ongoing — June 12 work tested on A32
10	Privacy Policy	Not started
11	Terms of Use	Not started
12	Open Source Credits page — include Open-Meteo attribution	Not started
13	Weather API licensing — awaiting Open-Meteo response	Email sent June 12
14	Play Store listing — description, screenshots, rating	Not started

18. Onboarding Flow — 5 Screens (Designed, Not Yet Built)

Blue color scheme locked in — matches Scout's eye color and visual identity. Designed by ChatGPT. Approved June 12.

IMPORTANT: Screen counter (e.g. '4 / 5') and progress dots must both be driven by the same variable when built in Android. Never hardcode the number in two places.

Screen	Title	Key Message
1 of 5	Welcome to Scout	'I'm Scout. Just say my name and I'll be listening.' No account required. Privacy-first. No subscriptions. E
2 of 5	Try Scout Free for 7 Days	\$9.99 one time. Never a subscription. Pay once. Own Scout forever.
3 of 5	This Is Just The Beginning	Future updates: calendar, news, expressions, languages. Scout is actively growing.
4 of 5	Your Privacy Matters	No account. Local features stay on device. You decide what Scout remembers.

5 of 5	Ready To Begin?	Scout's face fills the screen. 'This is just the beginning.' Start Using Scout button.
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The 'What's to come' section also lives in Settings → About Scout → Features & Future Plans so users can revisit it later.
Screen 1 'See & Recognize' description reads: 'I see faces, scenes, and more.' — not 'pets' which is not yet fully implemented.

19. Versioning System

Defined June 12, 2026. Scout uses semantic versioning with Scout-themed release names.

Type	Examples	When to Use
Launch	1.0	First public release.
Bug Fix	1.0.1, 1.0.2	Bug fixes, crash fixes, wake word fixes, stability improvements.
Feature	1.1, 1.2, 1.3	New features — calendar, memory improvements, expressions, language support.
Major	2.0, 3.0	Full mood system, major memory upgrades, hardware mode, big companion leaps.

Scout-Themed Version Names:

- Scout 1.0 — The Beginning
- Scout 1.1 — Growing Up
- Scout 1.2 — Learning More
- Scout 2.0 — A New Chapter

Google Play Version Codes:

Users see the Version Name. Google Play tracks the Version Code internally. Version Name 1.0 = Version Code 1. Version Name 1.0.1 = Version Code 2. Version Name 1.1 = Version Code 3.

Update Experience:

- After each update — Scout shows a one-time Welcome Back screen with what changed.
- Scout optionally speaks: 'I've learned a few new things since my last update.'
- About Scout → Update History shows every major version and improvements.
- Google Play 'What's New' updated with every release.

20. Legal & Website

Website:

- Current address: lippy-robotics.gt.tc
- Future domain options: lippyrobotics.com, scoutcompanion.com, meetscout.ai
- List website on: Facebook page, Google Play listing, About Scout screen, website footer.
- Add a 'What's New' or 'Scout Development Updates' page — shows Scout is actively growing.

Required for Launch:

Document	Priority	Notes
Privacy Policy	1 — Before launch	What data Scout collects. What stays on device. Gemini is optional. Contact info.
Terms of Use	2 — Near launch	Scout is as-is. No guarantees. Not medical/legal/financial advice. Keep it simple.

Open Source Credits	3 — At launch	llama.cpp, TinyLlama, Phi models, Android libraries. Many licenses require attribution.
Website link	1 — Before launch	lippy-robotics.gt.tc in About Scout, Google Play listing, Facebook page.

Inside the App — About Scout must contain:

- Version number
- Privacy Policy link
- Terms of Use link
- Open Source Licenses link
- Website: lippy-robotics.gt.tc
- Update History — every major version and what changed
- Contact / Get in Touch — opens email to lippyroboticslabs@gmail.com

Support email is ready: lippyroboticslabs@gmail.com. Auto-responder confirmed — sets expectations on response time, and asks for device model, Android version, and a short description for technical issues. This helps Patrick triage bug reports faster.

Goal: look professional and transparent, not write a 50-page legal manual. A clean footer and About screen with four links is enough for launch.

Weather API Licensing — Open Question:

Open-Meteo's free tier shows 'Commercial use: not permitted.' Scout is sold as a \$9.99 one-time purchase, which may classify as commercial use. An email was sent June 12 asking Open-Meteo to clarify for small one-time-purchase apps, and about attribution requirements under their CC BY 4.0 license.

- If Open-Meteo confirms free tier is fine for small apps — add their attribution to Open Source Credits and proceed.
- If Open-Meteo requires the \$29/month plan — a recurring API cost conflicts with Scout's no-subscriptions philosophy. Free, commercial-friendly weather API alternatives will be explored instead.
- Either way, Open-Meteo attribution belongs in Open Source Credits per their CC BY 4.0 license terms.